

1.

Overfitting of polynomial matching: We have shown that the predictor defined in Equation (2.3) leads to overfitting. While this predictor seems to be very unnatural, the goal of this exercise is to show that it can be described as a thresholded polynomial. That is, show that given a training set $S = \{(x_i, f(x_i))\}_{i=1}^m \subseteq (\mathbb{R}^d \times \{0, 1\})^m$, there exists a polynomial p_S such that $h_S(x) = 1$ if and only if $p_S(x) \geq 0$, where h_S is as defined in Equation (2.3). It follows that learning the class of all thresholded polynomials using the ERM rule may lead to overfitting.

The predictor defined in Equation (2.3) is the following:

$$h_S(x) = \begin{cases} y_i & \text{if } \exists i \in [m] \text{ s.t. } x_i = x \\ 0 & \text{otherwise.} \end{cases}$$

which basically just predicts based on successful training set predictions. In other words, we have that $h_S(x)$ is a perfect predictor on any point that was within the training data, and is 0 otherwise. To construct a polynomial p_S to emulate this, we need to ensure that p_S is nonnegative at every training point, and negative otherwise, as stated in the problem.

A very convenient expression that works for expressing this is in one-dimensional space is the polynomial $-(x - x_i)^2$, where we see that this is 0 when $x = x_i$, which satisfies the nonnegative requirement, and is clearly negative when $x \neq x_i$.

To extend this to d dimensions as with training set $\mathbb{R}^d$, we simply extend $x = (x_1, \dots, x_d)$ with $x_i = (x_{1i}, \dots, x_{di})$. With d dimensions, we can still emulate the behavior as with our prior one-dimensional building block expression, i.e. we can take the sum of all elements $(x_j - x_{ij})^2$ and this will give us our 0 sum when $x = x_i$, except now in d dimensions. This is fine as a one-hot indicator of whether $x = x_i$, and we can fix the negative behavior later.

It's possible (and very likely) that we have multiple training points, so we want to indicate a hit on any of these. To do this, we can take the product of this polynomial, that is,

$$\prod_{j=1}^d (\sum_{i=1}^m (x_j - x_{ij})^2)$$

which returns 0 if we hit on any $x = x_i$ for some i in the training set. We then fix that this indicates a positive value for all misses on the training set by adding a negative sign, which gives us

$$p_S = - \prod_{j=1}^d (\sum_{i=1}^m (x_j - x_{ij})^2)$$

Reflecting on the last statement of the problem, we can see that the predictor, in classifying every training example correctly, has training error of 0, and therefore could be selected by ERM because there is no empirical error less than 0, so it appears that this predictor is incredibly good by the ERM rule. The clear issue that arises from the selection of this predictor is that it will perform perfectly on any point within a training set, but given some finite number of training variables, it is possible (and very likely) that the true error will be higher as it evaluates data points outside of the training set. We can therefore see that the class of thresholded polynomials from the predictor selected using the ERM rule indeed can lead to overfitting.

2.

Let H be a class of binary classifiers over a domain X . Let D be an unknown distribution over X , and let f be the target hypothesis in H . Fix some $h \in H$. Show that the expected value of $L_S(h)$ over the choice of $S|_x$ equals $L_{(D,f)}(h)$, namely,

$$\mathbb{E}_{S|_x \sim D^m} [L_S(h)] = L_{(D,f)}(h).$$

The training error is defined as

$$L_S(h) = \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m}$$

and $L_{D,f}(h)$ is the true error. We can simplify this computationally by expressing success as Bernoulli variables, i.e. we count "1" for a successful classification, and "0" for an erroneous classification. We can therefore represent $L_S(h)$ as

$$L_S(h) = \frac{1}{m} \sum_{i=1}^m \{h(x_i) \neq f(x_i)\}$$

We take the overall expectation:

$$\mathbb{E}_{S|x \sim D^m} [L_S(h)] = \mathbb{E}_{S|x \sim D^m} \left[\frac{1}{m} \sum_{i=1}^m \{h(x_i) \neq f(x_i)\} \right]$$

By linearity of expectation:

$$= \frac{1}{m} \sum_{i=1}^m \mathbb{E}_{S|x \sim D^m} [\{h(x_i) \neq f(x_i)\}]$$

To evaluate the expectation component, we know that each x_i follows the distribution D , so we can evaluate this expectation of a singular instance i by taking the probability essentially just taking the probability that $h(x_i) \neq f(x_i)$ at that instance i , or that

$$\mathbb{E} [\{h(x_i) \neq f(x_i)\}] = \text{Prob}_{x_i \sim D} [h(x_i) \neq f(x_i)].$$

the latter of which we note is just the definition of the true error, $L_{D,f}(h)$. So we have that

$$\mathbb{E}_{S|x \sim D^m} [L_S(h)] = \frac{1}{m} \sum_{i=1}^m L_{D,f}(h)$$

$$\mathbb{E}_{S|x \sim D^m} [L_S(h)] = L_{D,f}(h)$$

exactly as we sought to find.

3.

Axis aligned rectangles: An axis aligned rectangle classifier in the plane is a classifier that assigns the value 1 to a point if and only if it is inside a certain rectangle...

1.

Let A be the algorithm that returns the smallest rectangle enclosing all positive examples in the training set. Show that A is an ERM.

2.

Show that if A receives a training set of size $\geq \frac{4\log(4/\delta)}{\epsilon}$ then, with probability of at least $1 - \delta$ it returns a hypothesis with error of at most ϵ . *Hint:* Fix some distribution $\mathcal{D}$ over $\mathcal{X}$, let $R^* = R(a_1^*, b_1^*, a_2^*, b_2^*)$ be the rectangle that generates the labels, and let f be the corresponding hypothesis. Let $a_1 \geq a_1^*$ be a number such that the probability mass (with respect to $\mathcal{D}$) of the rectangle $R_1 = R(a_1^*, a_1, a_2^*, b_2^*)$ is exactly $\epsilon/4$. Similarly, let b_1, a_2, b_2 be numbers such that the probability masses of the rectangles $R_2 = R(b_1, b_1^*, a_2^*, b_2^*)$, $R_3 = R(a_1^*, b_1^*, a_2^*, a_2)$, $R_4 = R(a_1^*, b_1^*, b_2, b_2^*)$ are all exactly $\epsilon/4$. Let $R(S)$ be the rectangle returned by A . See illustration in Figure 2.2.

- Show that $R(S) \subseteq R^*$.
- Show that if S contains (positive) examples in all of the rectangles R_1, R_2, R_3, R_4 , then the hypothesis returned by A has error of at most ϵ .
- For each $i \in \{1, \dots, 4\}$, upper bound the probability that S does not contain an example from R_i .
- Use the union bound to conclude the argument.

3.

Repeat the previous question for the class of axis aligned rectangles in $\mathbb{R}^d$.

4.

Show that the runtime of applying the algorithm A mentioned earlier is polynomial in $d, 1/\epsilon$, and in $\log(1/\delta)$.